PROCUREMENT PACKAGE:	ME063A - Cyclone Feed Pump
EPC **PROJECT NO.**	
EQUIPMENT NUMBER(S):	**EQUIPMENT NAME(S):**
F13-PP-401 F13-PP-402	CYCLONE FEED PUMP CYCLONE FEED PUMP SPARE (WET END ONLY)
Description:	SAG and Ball Mill discharge will be pumped via a single Cyclone Feed Pump from a common hopper to a single Cyclone cluster for classification.
Reference Specifications:	2000-F00-STS-GE-10002 – Site Specific Information 2000-F00-STS-EL-10001 – Electrical Equipment for Mechanical Packages 2000-F00-STS-EL-10002 – High Voltage Motor Specification 2000-F00-STS-EL-10014 – Preferred Equipment Suppliers for Electrical, Instrumentation and Controls Specification 2000-F13-DSH-EL-11001 – Cyclone Feed Pump VSD 2000-MEP-ME063-SCH1 – Vendor Document Schedule

SCOPE OF WORK

Work Included

Design, manufacture, assembly, and rotational testing the Cyclone Feed Pump including, but not limited to:

- Pump
  - One (1) heavy duty end suction centrifugal pump suitable for the operating conditions as nominated in the datasheet below.
  - Replaceable, adjustable liners.
  - Tungsten Carbide Cladding Wear Ring
  - Installed Rubber Suction Liner with spare Metal Suction Liner (c/w tungsten carbine cladding wear ring)
  - Seals - type as noted in datasheet below.
  - Easily interchangeable seal arrangement between gland and mechanical type seals.
  - Complete rotable spare wet end (F13-PP-402) complete with rotable support frame.
- Drive
  - Electric motor (direct drive) including Baumer HOG10 encoder included
  - High Speed couplings as detailed.
- Variable speed drive complete with step down transformer.
- Fabricated steel base frames, one for the pump and one for the motor
- Condition monitoring instrumentation and equipment, as quoted.
- 8 off Casing Liner wear sensors strategically located at high wearing areas.
- All ancillary items and special tools normally supplied with or required to adjust and maintain all components associated with the equipment.
- All fasteners, gaskets, packers, shims, and any other items necessary for site assembly.
- Certified lifting lugs as per Weir standard design.
- Stainless steel nameplates engraved with manufacturer, model, serial number, and equipment number.
- Surface treatment as per description in the datasheet
- Priced recommended spares list.

Specifications

- All equipment to be supplied in accordance with the above referenced specifications.

Documentation / Quality:

- Vendor data and documentation in accordance with agreed VDS.
- Shop assembly and testing as required to meet agreed documentation above.
- Pump performance and wear guarantee, including the following wear life:
  - Suction Liner - 26 weeks
  - Casing / Casing Liner - 52 weeks
  - Impeller - 26 weeks
  - Throat Bush - 26 weeks

Shipping:

- Packaging per Weir standard packing, suitable for road as sea transport.
- Delivery as nominated in the commercial documentation.

Work Excluded

Excluded from this scope of works shall be the following:

- Feed and discharge piping.
- Foundations and support structure outside battery limits.
- Hold-down bolts, nuts, and washers at structural steel.
- Motor starters
- Supply of all power, control, and instrumentation cabling outside battery limits.
- Lubricants and greases, other than initial fill quantities applied during shop assembly and /or preparation for shipment.
- Spare parts.
- Offloading the equipment at site.
- Installation and commissioning at site.

Battery Limits

The works covered by this datasheet extend up to, and include the following battery limits:

- Underside of base support frame.
- Pump suction and discharge flange connections.
- Seal water connection (if required).
- Electrical terminals of all supplied equipment:
    - Electric motor terminal boxes.
    - Instrument terminal boxes.

Design Requirements

General

- Adjustment of the wear ring / throatbush shall be made possible using hydraulic means from a remote location.
- Equipment will be operating in a harsh, outdoors environment handling abrasive slurry, thus, materials and surface treatments must be suitably selected to guarantee the longevity of the components.
- Design and materials for pump seals / glands to be suitable for the specified water quality.
- Base frames to be free draining to avoid pooling of liquid and supplied c/w SS shims and jacking bolts to facilitate alignment.
- Pump base frame to be designed and sized to retrofit a GIW MDX850 pump without major modification to the installed frame. The superintendent will provide a drawing of the GIW MDX850 for design.
- Battery operated auto grease system for each bearing shall be provided.

Guards:

- Guarding of the equipment to Weir standard in accordance with AS4024.
- All guards shall be capable of being removed without disturbing other parts of the equipment and where possible be a maximum weight of 30kg If this is not practical, supplier will seek approval from the Project, subsequent to approval lifting lugs or eye bolts shall be supplied, refer subsequent clauses on lifting requirements.
- Where guards interfere with access to lubrication points, stainless steel tubing and fixed remote grease nipples shall be provided so that lubrication can be undertaken without removing the guard and while the equipment is operating.
- Where any instruments that are inside a guarded area are required to be hand resettable – access from outside the guarding shall be provided to allow the instrument to be reset without risk to the operator.
- Guards shall have sufficient space for drive & component adjustment.

Lifting:

- Where practical all assemblies and components shall be provided with suitable lifting lugsas per Weir standard design to allow for easy off-loading and installation.
- Lifting lugs shall be stamped with the Working Load Limit (WLL) and provided with a Certificate of compliance stating that the lug conforms to relevant Australian Standards.

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA	NOTES
Site Data				
Site Location	-	WA		1
General Site Data	-	Refer to 2000-F00-STS-GE-10002 - Site Specific Information		1
Application Data				
Duty	-	Continuous		1
Location	-	Outdoors		1
Operating Hours per Day	hr/day	24		1
Operating Days per Week	days/week	7		1
Operating Weeks per Year	weeks/year	52		1
Utilisation	%	91.3		1
Design Life	years	30		1
Expected Ambient Operating Temperature Range	°C	Design for -5°C to 50°C		1

DUTY SPECIFICAITION	UOM	EPC DATA				NOTES
Process Requirements		**Minimum**	**Nominal**	**Max**	**Design**	
Flow Case Description		Single Stage SAG Mill	25.5 MTPA	28.35 MTPA	Design	1
Solids Mass Flow	tph	5,837	13,990	14,618	15,106	1
Liquid Mass Flow	tph	4,375	8,575	7,871	9,259	1
Slurry Flow Rate	m^3/h	6,102	12,838	12,397	13,862	1
Slurry SG	t/m^3	1.67	1.76	1.81	1.76	1
Slurry Temperature	°C	30	30	30	30	1
Solids Concentration	% w/w	57.2	62.0	65.0	62.0	1
Solids SG	-	2.90	2.90	2.90	2.90	1
Liquid SG	-	1.07	1.07	1.07	1.07	1
Particle Size (d_{50}) Refer to Table 1	µm	306	306	306	306	1
Particle Size (d_{80}) Refer to Table 1	µm	757	757	757	757	1
Slurry Viscosity	cP	5	5	5	5	1
Acidity	pH	5.5 – 8	5.5 – 8	5.5 – 8	5.5 – 8	1
Chlorides	ppm	86,243	86,243	86,243	86,243	1
Mechanical Requirements						
Suction Head	m slurry	1.6	1.6	1.6	1.6	1
TDH	m slurry	41.3	43.6	43.2	44.2	1
Head Ratio / Efficiency Ratio	-	0.96 / 0.92	0.96 / 0.92	0.96 / 0.92	0.96 / 0.92	3
Operating Speed at Flow Rate	rpm	207	226	223	233	2
Tip Speed at Flow Rate	m/s	26	28.4	28	29.3	2
Efficiency at Flow Rate	%	65.77 / 60.77	85.78 / 80.78	85.77 / 80.77	86.61 / 81.61	2
Absorbed Power at Flow Rate	kW	1896.98 / 2053.06	3391.08 / 3600.97	3349.15 / 3556.47	3676.41 / 3901.65	2
NPSH(a)	m fluid	6.31	5.66	5.55	5.57	1
NPSH(r)	m fluid	2.66	3.69	3.62	3.96	2
Seal Type (Mech/Gland/Exp)	-	Gland	Gland	Gland	Gland	3
Gland feed supply pressure	kPa	70-100 kPa above pump discharge pressure at all times including dead head	70-100 kPa above pump discharge pressure at all times including dead head	70-100 kPa above pump discharge pressure at all times including dead head	70-100 kPa above pump discharge pressure at all times including dead head	2

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA	NOTES
Pump Details				
Type	-	Horizontal Centrifugal Slurry Pump	Horizontal Centrifugal Slurry Pump	3
Quantity	#	1		1
Duty/Standby Arrangement	-	Duty		1
Manufacturer / Model	-	Weir MCR760 HF	Weir/Warman MCR760 HF	2
Performance Curve Number	-		EA7626	2
Pump Frame	-		M240	3
Pump Speed (Fixed / Variable)	-	Variable		1
Impeller Type	-	Closed	Closed	3
Number of Vanes	#		5	2
Installed Impeller Diameter	mm		2400	2

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA	NOTES
Maximum Impeller Diameter	mm		2400	2
Maximum allowed tip speed	m/s		38	2
Shut-Off Head at Design Operating Speed	m fluid		~54 m	2
Suction Diameter	DN		~965 mm	2
Discharge Diameter	DN		~762 mm	2
Discharge Orientation	-	Horizontal (Bottom)	Bottom Horizontal "G" will be requested (tbc by Eng)	3
Seal Make / Model	-		Gland Packed	2
Seal Water Quality	-		See M09 Manual attached	2
Maximum Particle Size	µm		100% of +250 mesh (60 µm)	2
Chloride Content	mg/L		Chloride (Cl-) ion max 1,000 ppm	2
Calcium Content	mg/L		Calcium Carbonate (CaCO3) max 10 ppm (mg/L)	2
Seal Water Flow Rate (New / Worn)	L/min		20 / 60	2
Seal Water Pressure Required	kPa(g)		35 kPag to 70 kPag above the pump discharge pressure at all times including dead head	2
Seal Materials	-		-	2
Materials of Construction				
Base Frame Material	-	AS 3678 Grade 250	E02 material based on AS 3678 Grade 250	1
Impeller Material	-	Weir A61 Material	A61 Hyperchrome	1
Casing Material	-		-	2
Casing Liner Material	-	Weir A68 Material	A68 Hyperchrome	1
Throat Bush Material	-	Weir A65 Material	A65 Hyperchrome	1
Shaft Material	-	Weir E43 Material	E43 steel	1
Shaft Sleeve Material	-	420 Stainless Steel coated with Tungsten Carbide (Weir J32 Material)	J32 or equivalent	2
Fasteners	-		-	2
Guarding	-		-	2
Stuffing Box	-	PTFE Kevlar Coated, Weir Q22 Material	Q22 or equivalent see MDS attached	2
Motor Details				
Motor Manufacturer / Type	-	Toshiba	Toshiba as selected by client	1
Installed Power	kW	4,800	4800 nominal	3
Motor IEC Frame Size / Motor Mounting Type	-		D1000 TBC/ IM1001 (IMB3) Foot mount	2
Starter Type (DOL/VSD/SS)	-	VSD	VSD	3
Number of Poles / Full Load Speed	# / rpm	22 / 266 rpm	12 / 237 tbc	3,4
Voltage / Frequency / Phase	V / Hz / ϕ	3300 / 50 / 3	3650 / 23.9/3 tbc	3
Enclosure (IP) Rating	-	66	56	3
Motor Rated for VSD Control	Yes / No	Yes		1
Hazardous Area Rated	Yes / No	No		1
Temperature Protection	Yes / No	Yes		1
Anti-Condensation Heater	Yes / No	Yes		1
Surface Protection/Paint Spec	-		OE spec Light Grey for Outdoor	2
Junction Box Details				
Make / Model	-		NA	2
Material of Construction	-	316 Stainless Steel		1
IP Rating	-	IP 65	IP56	1
Surface Protection				
Preparation	-		Abrasive Blast to AS1627.4 Class 2 1/2	2
Primer Type and DFT	µm		Epoxy Zinc Phosphate, 100 DFT	2
Top Coat Type, Colour and DFT	µm		Polyurethane, 50 DFT	2
Maintenance Information				
Change out time required for rotable spare wet end	hrs		As a thumb rule: 10-12 hours for clamshell, 6-8 hrs with keyed base bare shaft	2
Shipping Masses and Dimensions				
Complete Assembly	kg		~160000 tbc	2
Bare Shaft Pump	kg		~75000 tbc without cradle/baseplate	2

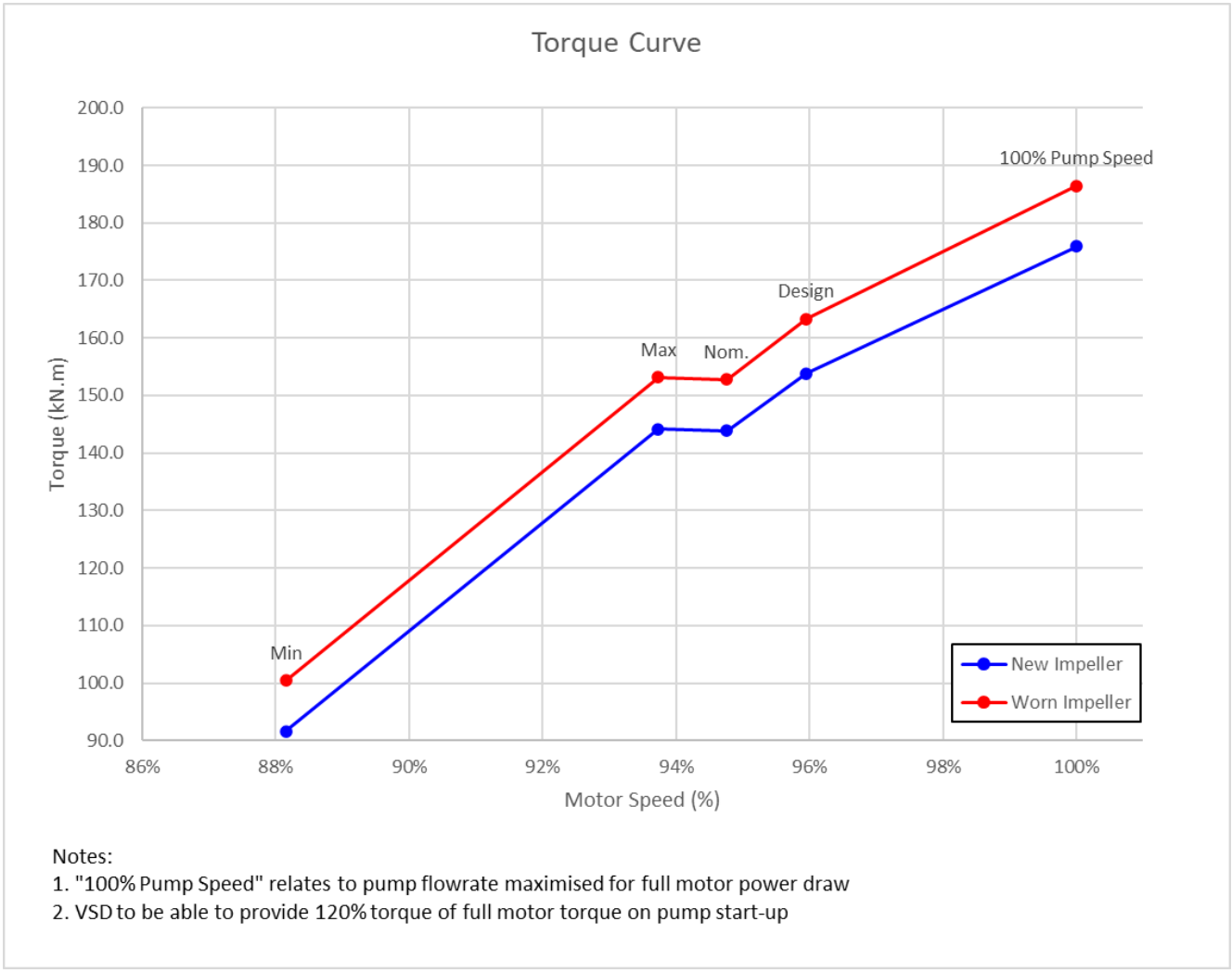
DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA	NOTES
List main components				
Cover Plate	kg		~10000	2
Impeller	kg		~10000	2
Liners	kg		~17000 metal liner	2
Motor	kg		~50000	2
Pre-Assembly State - Describe Site Assembly Required	-		Bare shaft pump on its base, motor dropped on motor baseplate , aligned & coupled with the pump	2

Notes:

1. Data supplied by EPC The Supplier shall comply with this data.
2. Data shall be provided by the Supplier.
3. Data proposed by EPC The Supplier shall confirm data or state alternative.
4. Optional pricing for a reduced pole motor and matching VSD may be offered by the Supplier. Motor and VSD pricing to be provided separately from the bare shaft pump.

Size (mm)	Passing (%)	Size (mm)	Passing (%)
350	100	6.7	99.6
250	100	4.75	99.2
190	100	3.35	98.3
165	100	2.36	96.5
140	100	1.7	93.6
125	100	1.18	88.8
100	100	0.85	83.1
90	100	0.6	74.8
63	100	0.425	63.0
45	100	0.3	49.3
31.5	100	0.212	37.6
25	100	0.15	29.2
22.4	100	0.106	23.1
16	100	0.075	18.4
13.2	100	0.053	14.7
9.5	99.8	0.038	11.9

Table 1 – Cyclone Feed PSD Consultant *Report 7555* Owner *PDC PSD Addendum Rev 1)*



Notes:
1. "100% Pump Speed" relates to pump flowrate maximised for full motor power draw
2. VSD to be able to provide 120% torque of full motor torque on pump start-up

Figure 1 – Torque vs Speed Curve (Vendor to provide updated curve to suit selected pump)